

Gabriela Shirkey, PhD

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EDUCATION

Ph.D., Geography, Environment & Spatial Science; Environmental Science & Public Policy
2023 *Michigan State University*
BS, Scientific & Technical Communication,
2013 *Michigan Technological University*

RESEARCH APPOINTMENTS

Postdoctoral Teaching Fellow, Grand Challenges Initiative 2023–Present
Supervisors: Drs. Joshua B. Fisher, Javier Espeleta, *Chapman University, CA*

Research Fellow, Landscape Ecology & Ecosystem Science 2017–2023
Supervisor: Dr. Jiquan Chen, *Michigan State University*

Lab Manager, Landscape Ecology & Ecosystem Science 2015–2017
Supervisor: Dr. Jiquan Chen, *Michigan State University*

Undergraduate Research Assistant 2013
Supervisor: Dr. Richelle Winkler, *Michigan Technological University*

RESEARCH GRANTS

NASA

1. CONNECT-SBG: Collaborative Nexus for Networking, Education, and Career Training in Surface Biology and Geology 2024-2026
NASA MOSAICS (\$300,000) | **PI**
2. Scaling the Structure, Composition, and Function of Terrestrial Ecosystem Biodiversity 2026-2029
NASA Biodiversity (\$1,000,000) | **CO-I**

Other grants

- Student travel award. AmeriFLUX Annual Meeting. 2019. (\$300) 2019
- College of Social Science Corey-Thompson Research Award. (\$5,000) 2019

PEER-REVIEWED PUBLICATIONS

Current H-Index 14 | Times cited >2000 | * indicates students mentored

29. **Shirkey, G.**, Goldsmith, G. R., Agatep, A.*, Camacho-Umaña, M. E., Dauber, G., Deering, C., Doughty, C., Espeleta, J. F., Fu, L., Hatch, H.*, Terrer, C., O'Connell, C. S., Yokoyama, G. & Fisher, J. B. (*in prep*). Tropical forest canopy height and estimated aboveground biomass increased in proximity to elevated canopy CO₂.
28. **Shirkey, G.**, Yokoyama, G., Fu, L., Agatep, A.*, Camacho-Umaña, M. E., Dauber, G., Deering, C., Doughty, C., Espeleta, J. F., Fu, L., Hatch, H.*, Jia, M., Nelson, K., Pivovarov, A., van Riper, S.*, Schwandner, F., Snyder, M., Terrer, C., Vargas Murillo, J. D.*, Zalman, C.,

- Goldsmith, G. R., O'Connell, C. S., & Fisher, J. B. (*in review*). Tropical stem efflux decreases under elevated canopy CO₂. *Scientific Reports*.
27. **Shirkey, G.**, Fisher, J.B., John, R., Saraf, S., Kolluru, V., Wang, X., Galvan, F.R., Fazli, S.* , Maharjan, S.* , Goldsmith, R.G. (*in review*). Biodiversity from the International Space Station: multiple lenses advance species richness monitoring. *npj Biodiversity*.
 26. Yokoyama, G., Fisher, J. B., **Shirkey, G.**, Agatep, A., Camacho-Umaña, M. E., Dauber, G., Doughty, C. E., Espeleta, J. F., Hatch, H., O'Connell, C. S., Pivovarov, A. L., van Riper, S., Terrer, C., Vargas Murillo, J. D., Zalman, C., & Goldsmith, G. R. (*in review*). Leaf photosynthetic trait variation in tropical forest trees exposed to long-term naturally elevated CO₂. *Global Change Biology*.
 25. Goldsmith, G.R., Martens, J., Patterson, M., Fisher, J.B., **Shirkey, G.**, Chang, E. (*in review*). Evapotranspiration in a large-scale enhanced rock weathering field trial and its effects on base cation concentrations in soil porewater export. *Environmental Science & Technology*.
 24. Fu, L.* , Yokoyama, **G.**, **Shirkey, G.**, Terrer, C., Camacho-Umaña, M. E., Fisher, R. A., Jia, M., Swann, A. L. S., Holm, J. A., Schwandner, F. M., O'Connell, C. S., Liu, A., Doughty, C. E., Zalman, C., Agatep, A., Hatch, H., Dauber, G., Nelson, K., Deering, C. D., Goldsmith, G. R., Fisher, J. B. (*in review*). A trade-off of carbon gain and water loss between wet and dry tropical forests under elevated CO₂. *PNAS*.
 23. Fazli, S.* , Maharjan, S.* , Wenzhao, Li, Rejoice, T., Fisher, J.B., Rejoice, T., Galvan, F.R., **Shirkey, G.**, El-Askary, H. (2025) Unraveling crop nitrogen-water dynamics with hyperspectral-thermal sensing in northern central valley, California. *Earth Systems and Environment*, 1-23.
 22. **Shirkey, G.**, Ancil, A., John, R., Kolluru, V., Mungai, L., Kashongwe, H., Cooper, L.T., Celik, I., Fisher, J.B. & Chen, J. (2025) Identifying opportunities for nature-based solutions with geospatialized life cycle assessments and fine-scale socioecological data. *Environmental Research Letters*, 20(1), 014023.
 21. Reed, D. E., Lei, C., Baule, W., **Shirkey, G.**, Chen, J., Czajkowski, K. P., & Ouyang, Z. (2024). Impacts of an urban density gradient on land-atmosphere turbulent heat fluxes across seasonal timescales. *Theoretical and Applied Climatology*, 1-10.
 20. Lei, C., Chen, J., Ibáñez, I., Sciusco, P., **Shirkey, G.**, Lei, M., Reich, P & Robertson, G. P. (2024). Albedo of crops as a nature-based climate solution to global warming. *Environmental Research Letters*, 19(8), 084032.
 19. Ravilla, A., **Shirkey, G.**, Chen, J., Jarchow, M., Stary, O., & Celik, I. (2024). Techno-economic and life cycle assessment of agrivoltaic system (AVS) designs. *Science of The Total Environment*, 912, 169274.
 18. Delwiche, K. B., Nelson, J., Kowalska, N., Moore, C. E., **Shirkey, G.**, Tarin, T., ... & Keenan, T. F. (2024). Charting the future of the FLUXNET network. *Bulletin of the American Meteorological Society*.
 17. **Shirkey, G.**, John, R., Chen, J., Kolluru, V., Amirkhiz, R. G., Marquart Pyatt, S. T., Cooper, L., Collins, M*. (2023) Land cover change and socioeconomic influences on terrestrial carbon production in an agroecosystem. *Landscape Ecology*. 38(12), 3825-3867.
 16. Belongueay, M.* , **Shirkey, G.**, Monteiro Lunardi, M., Rodriguez-Garcia, G., Sinha, P., Corkish, R., Steward, R.A., Ancil, A., Chen, J. & Celik, I. (2023) Photovoltaic systems through the lens of material-energy-water nexus. *Energies*. Accepted March 28, 2023.

15. **Shirkey, G.**, John, R., Chen, J., Sciusco, P., Abraha, M., Lei, C., Reed, D. (2022). Fine resolution remote sensing spectra improves estimates of gross primary production of croplands. *Agricultural and Forest Meteorology*, 326, 109175.
14. Xie, Y., Qi, J., Zhang, R., Jiao, X., **Shirkey, G.**, & Ren, S. (2022). Toward a carbon-neutral state: A carbon–energy–water nexus perspective of China’s coal power industry. *Energies*, 15(12), 4466.
13. Wu, S.R., **Shirkey, G.**, Celik, I., Shao, C., Chen, J. (2022). A review on the adoption of AI, BC, and IoT in sustainability research. *Sustainability* 14(13), 7851.
12. Sciusco, P., Chen, J., Giannico, V., Abraha, M., Lei, C., **Shirkey, G.**, Yuan, J., Robertson, G.P. (2022). Albedo-induced global warming impact at multiple temporal scales within an upper Midwest USA watershed. *Land*, 11(2), 283.
11. Chen, J., John, R., Shao, C., Ouyang, Z., Mack, E.A., Henebry, G.M., Dong, G., Allington, G.R.H., Pearson, A. L., Zhao, F., Roy, D.P., Fan, P., **Shirkey, G.**, Tian, L., Kussainova, M., Chen, J.Y., Reed, D., Abraha, M. (2021). Towards a single integrative metric on the dynamics of social-environmental systems. *Sustainability*, 13(20), 11246.
10. Abraha, M., Chen, J., Hamilton, S. K., Sciusco, P., Lei, C., **Shirkey, G.**, Yuan, J., Robertson, G. P. (2021). Albedo-induced global warming impact of Conservation Reserve Program grasslands converted to annual and perennial bioenergy crops. *Environmental Research Letters*, 16(8), 084059.
9. **Shirkey, G.**, Belongueay, M.*, Wu, S., Xiaoguang, M., Hassan, T., Anctil, A., Marquart-Pyatt, S., Stewart, R.A., Sinha, P., Corkish, R., Chen, J., Celik, I. (2021). An environmental and societal analysis of the US electrical energy industry based on the water-energy nexus. *Energies*, 14(9), 2633.
8. Sciusco, P., Chen, J., Abraha, M., Lei, C., Robertson, G. P., Laforteza, R., **Shirkey, G.**, Ouyang, Z., Zhang, R., John, R. (2020). Spatiotemporal variations of albedo in managed agricultural landscapes: inferences to global warming impacts (GWI). *Landscape Ecology*, 35(6), 1385-1402.
7. Dahlin, K. M., Akanga, D., Lombardozzi, D. L., Reed, D. E., **Shirkey, G.**, Lei, C., ... & Chen, J. (2020). Challenging a global land surface model in a local socio-environmental system. *Land*, 9(10), 398.
6. Fisher, J. B., Keenan, T. F., Buechner, C., **Shirkey, G.**, Perez-Quezada, J. F., Knox, S.H., Frank, J.M., Runkle, B.R.K, Bohrer, G. Once upon a time, in AmeriFlux. *Journal of Geophysical Research: Biogeosciences*, 126(1), e2020JG006148.
5. John, R., Chen, J., Giannico, V., Park, H., Xiao, J., **Shirkey, G.**, Ouyang, Z., Shao, C., Laforteza, R., Qi, J. (2018). Grassland canopy cover and aboveground biomass in Mongolia and Inner Mongolia: Spatiotemporal estimates and controlling factors. *Remote Sensing of Environment*, 213, 34-48.
4. Sat Gungor, B., Chen, J., Wu, S. R., Zhou, P., & **Shirkey, G.** (2018). Does plant knowledge within urban forests and parks directly influence visitor pro-environmental behaviors. *Forests*, 9(4), 171.
3. Fu, Z., Stoy, P.C., Luo, Y., Chen, J., Sun, J., Montagnani, L., Wohlfahrt, G., Rahman, A.F., Rambal, S., Bernhofer, C., Wang, J., **Shirkey, G.**, Niu, S. (2017). Climate controls over the net carbon uptake period and amplitude of net ecosystem production in temperate and boreal ecosystems. *Agricultural and Forest Meteorology*, 243, 9-18.

2. Fan, Y., Chen, J., **Shirkey, G.**, John, R., Wu, S.R, Park, H., Shao, C. (2016). Applications of Structural Equation Modeling (SEM) in Ecological Studies: An Updated Review. *Ecological Processes*, 5(1), 1-12
1. Lind-Riehl, J., Jeltema, S., Morrison, M., **Shirkey, G.**, Mayer, A., Rouleau, M., Winkler, R. (2015). Family legacies and community networks shape private forest management in the western Upper Peninsula of Michigan (USA). *Land Use Policy*, 45, 95–102.

FELLOWSHIPS & RECOGNITION

- Postdoctoral Teaching Fellowship on Experiential Learning—(2023–2026)
- FLUXNET 2023 Service Award
- National Science Foundation Graduate Research Fellow—\$34000 (2019–2024)
- MSU Environmental Science & Public Policy Outstanding Student Paper Award—\$500
- University COGS Disciplinary Leadership Award—\$1,000 (2022)
- MSU Environmental Science & Public Policy Doctoral Recruiting Fellowship (2018)
- “Women in STEM”. *The Anchor Magazine*. National Sorority Alpha Sigma Tau (2018)
- MSU Academic Achievement Graduate Assistantship (2017)
- Michigan Technological University Traditions of Giving Scholarship (2012)

TEACHING ACTIVITIES

<u>Instructor of Record</u>	2023–Present
<i>Chapman University</i>	
▪ ENV 241: Observing Earth from Above ▪ FFC100-B: Foundational concepts in STEM applications to solve societal problems ▪ GCI-150: Creating a scientific approach to a societal challenge ▪ GCI-200: Introduction to the research process and teamwork ▪ GCI-250: Science communication and poster showcase	
<u>Curriculum Developer & Coordinator</u>	2024–Present
<i>Chapman University, Grand Challenges in Science & Engineering</i>	
▪ GCI-150: Creating a scientific approach to a societal challenge ▪ GCI-200: Introduction to the research process and teamwork	
<i>Chapman University, Schmid College of Science and Technology</i>	
▪ ENV 231: Observing Earth from Above: Using NASA Satellite Data for Real-Time Environmental Monitoring	2026
<u>Curriculum Developer & Coordinator – The Graduate School</u>	2022
<i>Michigan State University</i>	
▪ PILOT-Pursuing Intentional Learning in Organizations Together	
<u>Teaching Assistantship – Geography, Environment & Spatial Sciences</u>	
<i>Michigan State University</i>	
▪ GEO 221: Introduction to Geographic Information	2022

- GEO 203v: Introduction to Meteorology

2023

Training and professional development in teaching

2023–Present

Center for Excellence in Teaching and Learning, Chapman University

- Course Design Institute (3-day workshop)
- Every term needs a plan: facilitating student growth through feedback
- Organizing productivity: building student agency
- Strategies for Designing Effective Rubrics

Center for the Improvement of Mentored Experiences in Research (CIMER)

- Entering Mentoring

UNDERGRADUATES MENTORED

21. Gil Lopez Garcia, Schmid College of Sci. & Tech., Env. Sci. & Policy, Chapman University
20. Sam Kuhn, Schmid College of Sci. & Tech., Env. Sci. & Policy, Chapman University
19. Quin Meiseles, Schmid College of Sci. & Tech., Env. Sci. & Policy, Chapman University
18. Juliet Holmboe, Dodge College of Film and Media, Chapman University
17. Joshua Fury, Fowler School of Eng., Data Science, Chapman University
16. Fares Alhezaimi, Fowler School of Eng., Data Science, Chapman University
15. Mona Patterson, Dodge College of Film and Media, Chapman University
14. Maisy Runkel, Schmid College of Sci. & Tech., Env. Sci. & Policy, Chapman University
13. Ryan Joshi, Fowler School of Eng., Data Science, Chapman University
12. Kyla Monique Cabrera, Fowler School of Eng., Data Science, Chapman University
11. Micah Kim, Schmid College of Sci. & Tech., Env. Sci. & Policy, Chapman University
10. Holland Hatch, Schmid College of Sci. & Tech., Env. Sci. & Policy, Chapman University
9. Ashley Agatep, Schmid College of Sci. & Tech., Env. Sci. & Policy, Chapman University
8. Juliette Vasquez, Schmid College of Sci. & Tech., Env. Sci. & Policy, Chapman University
7. Ava Cison, Schmid College of Sci. & Tech., Env. Sci. & Policy, Chapman University
6. Megan Belongay, Dept. Engineering Physics, University of Wisconsin–Platteville
5. Michael Collins, Dept. Environment and Sustainability, Michigan State University
4. Kelsey Watson, Dept. Environmental Studies and Sustainability, Michigan State University
3. Kaitlyn O'Brien, Dept. Community Sustainability, Michigan State University
2. Kaylee Peterson, Dept. Community Sustainability, Michigan State University
1. Ezequiel Mussambe, Dept. Geography, Environment & Spatial Sciences, Michigan State University

GRADUATES MENTORED

1. Lianlei Fu, Fowler School of Eng., Data Science, Chapman University
2. Sharhyar Fazli, Fowler School of Eng., Data Science, Chapman University
3. Surendra Maharjan, Fowler School of Eng., Data Science, Chapman University

POSTER PRESENTATIONS

** indicates students mentored*

22. **Shirkey, G.**, Alhezaimi, F.*, Goldsmith, G. R., Fisher, J. B. (2026). Scaling the structure, composition, & function of terrestrial ecosystem biodiversity. *NASA Biosphere*. 07/09/26.
21. **Shirkey, G.**, Agatep, A.*, Camacho Umaña, M., Dauber, G.*, Doughty, C.*, Espeleta, J.F., Fu, L.*, Goldsmith, G.R., Hatch, H.*, Pivovarovoff, A., Riper, S*, Snyder, M., Terrer, C., Vargas Murillo, J., Zalman, C., O'Connell, C., Yokoyama, G., Fisher, J.B. ELEVATE: CO₂ Fertilization Experiment in a Costa Rican Neotropical Forest. *American Geophysical Union*. 12/17/25.
20. Joshi, R.*, Stonebraker, R., **Shirkey, G.**, Fisher, J.B. Drought detection model for New Mexico. *ECOSTRESS Annual Meeting*. 9/23/25.
19. Kim, M.*, Luis, K., **Shirkey, G.**, Tanner, R., Environmental effects on coastal housing via remote sensing. *ECOSTRESS Annual Meeting*. 9/23/25.
18. Fu, L.*, Stonebraker, R., Fisher, J.B., Shirkey, G. Pushing the limits: 3m resolution evapotranspiration estimates using PT-JPL. *ECOSTRESS Annual Meeting*. 9/23/25.
17. Cabrera, K.*, Luis, K., Tanner R., **Shirkey, G.** Spectral variability from space: How well do satellites monitor intertidal zones? *ECOSTRESS Annual Meeting*. 9/23/25.
16. Alhezaimi F.*, **Shirkey, G.**, Joshi, R., Fisher, J.B. Comparing machine learning models for plant biodiversity using NEON, EMIT, and ECOSTRESS data. *ECOSTRESS Annual Meeting*. 9/23/25.
15. **Shirkey, G.**, Yokoyama, G., Agatep, A.*, Camacho Umaña, M., Dauber, G.*, Doughty, C.*, Espeleta, J.F., Fu, L.*, Goldsmith, G.R., Hatch, H.*, Pivovarovoff, A., Riper, S*, Snyder, M., Terrer, C., Vargas Murillo, J., Zalman, C., O'Connell, C., Fisher, J.B. Effects of long-term elevated CO₂ on tropical tree & soil CO₂ and CH₄ fluxes. *Ecological Society of America Annual Meeting*. 8/13/25.
14. **Shirkey, G.**, Yokoyama, G., Agatep, A.*, Camacho Umaña, M., Dauber, G.*, Doughty, C.*, Espeleta, J.F., Fu, L.*, Goldsmith, G.R., Hatch, H.*, Pivovarovoff, A., Riper, S*, Snyder, M., Terrer, C., Vargas Murillo, J., Zalman, C., O'Connell, C., Fisher, J.B. Effects of long-term elevated CO₂ on tropical tree & soil CO₂ and CH₄ fluxes. *Association of Tropical Biology and Conservation Annual Meeting*. Oaxaca, MX. 6/30/25.
13. Cabrera, K.*, Kim, M.*, **Shirkey, G.**, Luis, K., Stoneman, A.*, Wakamatsu, E.*, Tanner, R. Intertidal spectral-taxonomic relationships in Southern California. *Surface Biology and Geology (SBG) Science & Applications Technical Interchange Meeting (TIM)*. Virtual. 5/22/2025.
12. **Shirkey, G.**, John, R., Fazli, S.*, Maharjan, S.*, Romero, F.*, Saraf., S.*, Fisher, J. (2024) An ISS Synergy to Detect Biodiversity from Above. American Geophysical Union, Washington D.C. 12/11/24.
11. **Shirkey, G.**, Chen, J., John, R., Anctil, A., Celik, I., Kashongwe, H., Kolluru, V., Cooper, L. (2023) Carbon offset potentials of agroecosystems: a spatialized life-cycle assessment approach. American Geophysical Union, San Francisco, CA. 12/17/23.

10. **Shirkey, G.**, John, R., Chen, J., Sciusco, P., Abraha, M., Lei, C., Reed, D. (2023). Fine resolution remote sensing spectra improves estimates of gross primary production of croplands. *FLUXNET 2023 Annual Meeting*. Brno, Czech Republic. 7/17-20/23.
9. **Shirkey, G.**, John, R., Chen, J., Kolluru, V., Amirkhiz, R. G., (2022). Land cover land use change and socioeconomic influences on terrestrial carbon production in a southwest Michigan watershed. *International Association of Landscape Ecologists North America Annual Meeting*. Riverside, California. 3/20-23/23.
8. **Shirkey, G.**, John, R., Chen, J., Kolluru, V., Amirkhiz, R. G., (2022). Land cover land use change and socioeconomic influences on terrestrial carbon production in a southwest Michigan watershed. *American Geophysical Union Fall Meeting*. Chicago, Illinois. 12/12-16/22.
7. **Shirkey, G.**, John, R., Chen, J., Scuisco, P., Abraha, M., Lei, C., Reed, D. (2022). Fine resolution remote sensing spectra improves estimates of gross primary production of croplands. *AmeriFlux Annual Meeting*. University of Michigan Biological Station, Michigan. 09/09/22.
6. **Shirkey, G.**, John, R., Scuisco, P., Dahlin, K., Abraha, M., Lei, C., Reed, D., Chen, J. (2021). Modeling of gross primary production compared across scales in managed agricultural-grassland landscapes. *Specnet Fall Meeting*. Virtual. 10/29/21.
5. **Shirkey, G.**, John, R., Chen, J., Abraha, M., Scuisco, P., Reed, D. (2020). Evaluation of satellite-derived gross primary production products across managed Midwest landscapes. *AmeriFlux 2020 Annual Meeting*. Virtual. 10/06-08/20.
4. **Shirkey, G.**, Reed, D., Abraha, M., Peterson, K., Scuisco, P., Watson, K., Cooper, L., Chen, J., Dahlin, K. (2018). Transposing Socioeconomic and Qualitative Research into a Historical, Landscape-scale Assessment of Carbon Stock in Agricultural-Forestland. *American Geophysical Union (AGU) fall meeting*. Washington D.C., MD. 12/14/18.
3. **Shirkey, G.**, Scuisco, P., John, R., Reed, David., O'Brien, K., Lei, C., Cooper, L., Chen, J. (2018). Integrating historical land cover and land management in Michigan's Kalamazoo Watershed: a story of carbon flux impact. Fate of the Earth Symposium. Environmental Science and Policy Program, Michigan State University, MI. 3/24/18.
2. **Shirkey, G.** (2018). Implementing an agricultural Life Cycle Assessment (LCA) at the regional scale in southwest Michigan. *Global Change and Agroecosystems: Challenges and Opportunities*. Michigan State University, East Lansing, MI. 1/15/18.
1. **Shirkey, G.**, Chen, J., John, R., Park, H., Liang, M., and Giannico, V. (2017). Contributions of tree mortality to C accumulation in an oak savanna system: a decadal summary. *Association of American Geographers (AAG)*. Boston, MA. 4/7/17.

Reports & other publications

- Bose, R., **Shirkey, G.**, Wakeham, T., Karvakko, C., Kreuze, A., Louie, E., Morrison, M., Sanchez-Gonzalez, M., Geisler, D., Winkler, R. (2013). *Exploring the Feasibility of Minewater Geothermal in Calumet*. Michigan Technological University. Community Report.

ORAL PRESENTATIONS

7. **Shirkey, G.** (2026). Drone-based LiDAR vegetation structure in wet and seasonally dry tropical forests under long-term elevated CO₂. *Ecological Society of America*. 07/30/2026.
6. **Shirkey, G.** (2025). CONNECT-SBG: Collaborative Nexus for Networking, Education, and Career Training in Surface Biology and Geology. *American Geophysical Union*. 12/16/2025.

5. **Shirkey, G.** (2025). Identifying opportunities for nature-based solutions with geospatialized life cycle assessments and fine-scale socioecological data. *American Association of Geographers*. Detroit, MI. 3/24-28/25.
4. **Shirkey, G.** (2023). Fine resolution remote sensing spectra improves estimates of gross primary production of croplands. *International Association of Landscape Ecologists North America Annual Meeting*. Riverside, CA. 3/20-23/23.
3. **Shirkey, G.** (2022). Michigan agricultural landscapes: Socio-ecological contributions to climate change. *ESPP 2022 Spring Research Colloquium Marathon*. 3/23/22. Virtual.
2. **Shirkey, G.** (2021). An environmental and societal analysis of the US electrical energy industry based on the Water-Energy Nexus. *FEWSUS Student Spring 2021 Symposium*. 4/27/21. Virtual.
1. **Shirkey, G.**, Scuisco, P., Mussambe, E., Lei, C., John, R., Portelli, R., & Chen, J. (2019). Re-imagining the past: incorporating historical aerial photographs in landscape scale analysis. *ESPP Research Symposium*. East Lansing, MI. 10/28/19.

Invited Talks

- **Shirkey, G.** (2025). CONNECT-SBG: Collaborative Nexus for Networking, Education, and Career Training in Surface Biology and Geology. *ECOSTRESS Annual Meeting*. 9/2025.
- **Shirkey, G.** (2025) Scaling socioeconomic data with ecosystem processes for evidence-based climate policy. *NASA JPL Carbon Club*. 2/2025.

SESSIONS & WORKSHOPS ORGANIZED

7. NASA Earth Science Coding and Career Workshop, Chapman University (April 2025)
6. FLUXNET Annual Meeting, Co-chair, Brno, Czech Republic (July 2023)
5. “FLUXNET ECN Spring Workshop”, virtual (Apr. 2021)
4. FLUXNET Annual Meeting 2020, Co-chair, virtual (Sept. 2020)
3. “SESYNC Pursuit Workshop: Energy-Water Nexus Analysis of the Solar Energy Industry”, virtual (Nov. 2020)
2. “GEE 101: Code your own site description. Coding workshop”, AmeriFlux Annual Meeting. Boulder, CO, USA. (2019)
1. “Land Management in the Earth System: Measurements and Models”. Session. American Geophysical Union Fall Meeting (2018)

PROFESSIONAL SERVICE

- FLUXNET/AmeriFlux Early Career Network (ECN) Organizer (Jun. 2019–2023)
- Reviewer: *New Phytologist*– Wiley & Sons (2025-Present)
- Reviewer: *Science of the Total Environment*– Elsevier (2022-Present)
- Reviewer: *LAND* – MDPI (2022-2023)

OUTREACH & PUBLIC ENGAGEMENT

- Organizer, “Intellectual Will and Branding Workshop with Alumna *Lisa-Marie Pierre*”, Supporting Women in Geography (May 2022)

- Organizer, "Raising Awareness of Neurodiversity & Space", Supporting Women in Geography, Michigan State University (Mar. 2022)
- Organizer and committee member, "Graduate letter of demands", a movement to improve grad-faculty relations, education and DEI at MSU Department of Geography (2020-2021)
- Organizer, "Open forum for Department of Geography on the impact at MSU from Larry Nassar" (Feb. 2018)
- Battle Creek Math + Science Center, Battle Creek, MI. Coordinated lesson material on ecology and micrometeorological instrumentation (2017-2018)
- Girls in Math + Science, Michigan State University, MI. Youth educator (2017, 2018)
- SciFest, Michigan State University. Logistics volunteer (2017, 2018)
- Michigan Inland Lakes Partnership, Michigan State University. Marketing, education (2016)
- Huron River Watershed Council, Ann Arbor, MI, data collection, youth leadership (2014-2016)
- Noche de Ciencias, Society of Hispanic Professional Engineers (SHPE) youth education in Grand Rapids and Detroit public schools (2012-2014)

MEMBERSHIP IN PROFESSIONAL SOCIETIES & NETWORKS

- FLUXNET, AmeriFlux (2019–Present)
- Supporting Women in Geography (2017–2023)
- Member, Gamma Theta Upsilon – Honorary Society in Geography (2019–Present)
- Member, Ecological Society of America (2017–Present)
- Member, American Geophysical Union (2017–Present)
- Member, American Association of Geographers (2017–Present)

PROFESSIONAL DEVELOPMENT & RESEARCH TRAINING

- Spectral Ecology Summer School (SPEC School). Pembroke, VA. (2025)
- FluxCourse. University of Colorado, Mountain Research Station. Boulder, CO. (2022)
- New Advances in Land Carbon Cycle Modeling. Northern Arizona University, Center for Ecosystem Science and Society, ECOLAB. Virtual. (2022)
- MSU Dialogues. Diversity training on the topic of race. Office for Inclusion and Intercultural Initiatives, Michigan State University, MI. (2018)
- NEON Surface Atmosphere Exchange Workshop. American Geophysical Union Fall meeting. Washington, D.C., MD. (2018)
- Bayesian Coding Workshop. Center for Global Change and Earth Observations, Michigan State University. (2018)
- LCLUC Synthesis Workshop: Ecosystem-Society Interactions on a Changing Mongolian Plateau. University of Michigan. (2016)
- Future Earth Synthesis Workshop. Michigan State University, Center for Global Change and Earth Observations. (2015)

- Regional Leadership Development Conference. Society of Hispanic Professional Engineers. Dayton, OH. (2013)
- Society of Hispanic Professional Engineers National Conference. Dallas, TX. (2012)

FIELD EXPERIENCE

International & domestic ecological sampling in grasslands and tropics – Support (2015-Present)

- Perform research with an international team in remote locations of Mongolia & Costa Rica
- Collect and synthesize datasets including aerial LiDAR, soil, litter, root core, leaf spectra, leaf traits, and other ecosystem characteristics
- Collaborate with international scholars and stakeholders to build long-term relationships

Eddy Covariance Tower Maintenance – Support (2017-2023)

- Identify stakeholders willing to host four new research towers for extended periods of time
- Contact construction crews, purchase hardware, and calibrate sensors
- Construct and maintain the tower, regularly check data acquisition

Stakeholder surveys of Michigan Centennial Farmers – Lead (2018-2019)

- Collaborate with Michigan Centennial Farmers Historical Society to recruit stakeholders
- Conducted 113 mail-in surveys, 32 in-person & supervised undergraduate researcher

Forest Inventory Analysis – Lead (2017-2022)

- Identify and install research plots according to the forest inventory analysis protocol
- Supervise the collection and data management of samples

Stakeholder surveys of non-industrial, private forest owners – Lead (2013)

- Recruit, record and transcribe interviews for analysis in HyperResearch software
- Identify opinion leaders, maintain database of stakeholders

ADDITIONAL SKILLS

- Language: English (Fluent), Spanish (Intermediate)
- Geospatial: ArcGIS, ArcPro, Google Earth Engine, TerrSet, ENVI, ERDAS
- Technical: Geospatial analysis, land cover land use change, ecosystem modeling, structural equation modeling, life cycle assessment, HyperResearch, EdiRe, REddyProc gapfilling, R/R Studio, Python, LoggerNet, data cleaning and management, statistical analysis
- Qualitative: Mail and in-person surveys, transcription, codebook and sentiment analysis
- Instrumentation: LI-7500 open-path infrared gas analyzer, CSAT3 three-dimensional sonic anemometer, CNR1, HMP45C, CR5000 datalogger

REFERENCES

Research

- Dr. Joshua B. Fisher (Present Mentor):
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Teaching

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Professional Service

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